A population-level divide in human sensitivity to musical harmony

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Author Note. All data and materials, including a fully reproducible version of this manuscript, are available at https://github.com/themusiclab/tone-scrambles. For assistance, contact S.W.

This study identifies a new class of subjects with a musically important type of auditory sensitivity called MM-sensitivity. Participants (N = 59,897) in a gamified experiment listened to random sequences of tones from either a major or a minor triad and strove to classify what they heard. Performance was strongly bimodal: a minority of "super-listeners" (with high MM-sensitivity) performed near ceiling, while most subjects ("non-supers") performed near chance. A pre-registered direct replication (N = 25,758) confirmed these findings. The heat map relating years-of-music-lessons to performance reveals (1) many super-listeners who never had lessons, and (2) many non-supers whose years of lessons failed to increase their MM-sensitivity. Whether lessons increase MM-sensitivity in some subjects remains unknown. Super-listeners tend to take many more years of lessons than non-supers, suggesting that MM-sensitivity facilitates musical achievement. We argue that super-listeners may have heightened sensitivity to the qualities conferred to music by variations in scale.

1. Introduction

This work establishes the existence of a special category of subjects that we call "super-listeners". Super-listeners reliably report greater musical achievement than other subjects, and they are easy to identify using a simple perceptual task. This task requires no musical background, nor does it require subjective assessments about musical passages; it is a simple categorization task in which each stimulus is either a G-major or G-minor random arpeggio. The difference in performance is stark: super-listeners perform near perfectly, but most subjects (non-supers) do hardly better than guessing. That such a sharp divide separates super-listeners from the majority, and that this divide predicts musical accomplishment suggests that the latent cause of super-listenerhood is importantly connected to music perception and, perhaps, to musical culture more generally.

We initially tested 59,897 subjects from around the world and found that super-listeners form a distinct category from non-supers. We then confirmed this result in a preregistered direct replication across a separate sample of 25,758 subjects.

1.1 Normal musical abilities

Nearly all subjects possess basic musical abilities. Two of the most important are the abilities to organize sound hierarchically based on its pitch variations and on its rhythmic variations (Mehr, 2025; Patel, 2008; Peretz & Morais, 1989). In the realm of pitch, as a melody unfolds, we hear certain pitches as more stable or central and others as leading away or back toward them (Janata & Reisberg, 1988; Krumhansl & Shepard,

1979; Krumhansl & Kessler, 1982; Krumhansl, 2004; Large et al., 2016); in the realm of rhythm, we naturally group beats into patterns of strong and weak pulses, forming meters and nested structures (Jones & Boltz, 1989; London, 2012; Large et al., 2009; Lerdahl & Jackendoff, 1983). People are also sensitive to relative pitch (recognizing a melody even when it is transposed) (Dowling, 1978; Krumhansl, 2001; Shepard, 1964), to temporal regularity (detecting a steady beat and noticing deviations from it) (Grahn & Brett, 2007; Iversen & Patel, 2008), and to patterns of tension and release over time (Huron, 2006; Koelsch, 2014; Krumhansl, 1996; Lerdahl, 2005; Meyer, 1962). Even infants demonstrate many of these abilities (Bianco et al., 2026; Hannon & Johnson, 2005; Trehub & Hannon, 2006; Winkler et al., 2009), suggesting they are deeply rooted in human biology. Together, these basic skills allow all subjects (except those with specific musical deficits), across different cultures, to make sense of music without any training (Mehr et al., 2019).

Although super-listeners and non-supers alike possess these basic musical capabilities, the fact that super-listeners tend to attain higher levels of musical competence suggests that they possess some musically important capability that non-supers lack. A basic question motivating this work is: what is the nature of this capability?

1.2 Past tone-scramble studies

Previous experiments have revealed bimodal performance in the tone-scramble task. However, the current study shines a brighter light on this phenomenon by sampling vastly more subjects from more diverse backgrounds than prior studies.

Within the Western musical tradition, subjects tend to characterize in the major and minor scales as sounding "happy" and "sad" respectively (e.g., Dalla Bella et al., 2001; Bonetti & Costa, 2019; Cunningham & Sterling, 1988; Gerardi & Gerken, 1995; Gagnon & Peretz, 2003; Heinlein, 1928; Hevner, 1935; Kastner & Crowder, 1990; Peretz et al., 1998; Temperley & Tan, 2013).[1] However, the emotional effect of a piece also depends on other features (e.g., tempo, rhythm, timbre, dynamics) of the music.

Chubb et al. (2013) introduced tone-scrambles to isolate effects of scale from other aspects of musical structure. These stimuli are randomly ordered sequences of pure tones, all equal in duration and loudness. In Chubb et al. (2013), each tone-scramble comprised notes drawn either from a G-major or a G-minor triad, and the task was to categorize it accordingly. This experiment revealed dramatic individual differences: roughly one-third of subjects (super-listeners) performed near ceiling; the rest (non-supers) performed near chance. Subsequent experiments yielded similar results (Dean & Chubb, 2017; Mednicoff et al., 2018; Ho & Chubb, 2020; Ho et al., 2022).

1.3 Do non-supers suffer from a labeling problem?

For all subjects except those with perfect pitch, it is difficult to identify notes presented in isolation. However, nearly all subjects can discern whether two notes presented sequentially are different. Perhaps, analogously, non-supers have difficulty identifying individual major and minor tone-scrambles but can discern whether tone-scrambles differ if they are presented sequentially. To investigate this possibility, we use a novel same-different task in which the subject hears two tone-scrambles sequentially and must guess whether they are the same type or different types. If non-supers suffer from a labeling problem, they should perform better in this task than in the original tone-scramble task.

1.4 Does super-listenerhood depend on early-life language exposure?

Do native speakers of tonal languages (i.e., languages that assign different meanings to words with identical syllables but different pitch contours (Pike, 1948; Hulst et al., 2010)) perform better on tone-scramble tasks than other subjects? Several previous results raise this possibility. For example, perfect pitch is more prevalent among native tonal-language speakers (Deutsch et al., 2004, 2006), and native tonal-language speakers

[1] These associations may not be shared across all cultures (Athanosopoulos et al., 2021; Smit et al., 2022).

perform better than other subjects in relative pitch tasks (Hove et al., 2010) and melodic discrimination tasks (Liu et al., 2023). We will use a self-report item querying native language to address this question.

2. Methods

2.1 Participants

Participants were visitors to the citizen-science website https://themusiclab.org who followed a link to the "Are You a Super-Listener?" game. They discovered the website and game by word-of-mouth (e.g., through postings on Reddit) and, as in many gamified experiments (see Long et al. (2023)), were provided with a visualization of their performance rather than being compensated for their participation. All participants gave informed consent under an ethics protocol approved by the Yale University Human Research Protection Program (protocol 2000033433).

We first analyzed data from 59,897 people who participated between February 22, 2021, and April 4, 2022. These data were drawn from an initial sample of 149,169 website visitors, from which we excluded participants in the same fashion as prior studies run on the same website (Liu et al., 2023); see SI Sec. S1.1 for a list of exclusion criteria.

As found in previous laboratory studies, performance in the tone-scramble task was bimodally distributed. We then preregistered our initial analyses and results and proceeded to analyze data collected after April 4, 2022, until February 14, 2025. These data represented 25,758 additional participants, excluding some visitors based on the same criteria as above. Altogether, the present sample size is 85,655.

2.2 The "Are You a Super-Listener?" game.

The game comprised three tone-scramble tasks: the major-minor categorization task, the major-minor same-different task, and the wildcard categorization task. The order of the major-minor categorization and same-different tasks was randomly determined for each subject. The wildcard categorization task came last.

At the start of each task, the subject viewed text instructions and performed a brief training sequence comprising four labeled examples and four practice trials. The subject could repeat this sequence as many times as desired.

2.2.1 Tone-scramble construction

In all tasks, a tone-scramble comprised 12 consecutive "pips", each a 65-ms pure tone windowed by a raised cosine function with 22.5 ms rise and decay times; each tone-scramble lasted 780 ms. All pips had equal amplitude, and volume was adjusted by each subject to a comfortable level prior to the experiment. The notes in a given tone-scramble were presented in a random order. The stimuli were generated at trial time and played using the Web Audio API (Adenot et al., 2021). The notes of the pips in all stimuli were drawn from the equal-tempered chromatic scale from G_5 to G_6. Every tone-scramble in each of the 9 different tasks included three copies of each of the notes, G_5, D_6, and G_6. This ensured that a tonic of G was well-established on every trial in every task. Any given tone-scramble also included 3 pips of an additional "target" note.

2.2.2 The major-minor categorization (MMC) task.

Each stimulus was a single tone-scramble. In major (minor) tone-scrambles, the target note was $B\natural_5$ ($B\flat_5$). The participant indicated which type they heard and received feedback. This task comprised 20 trials including 10 major and 10 minor stimuli presented in random order.

2.2.3 The major-minor same-different (MMSD) task

Each stimulus comprised two tone-scrambles separated by a 300 ms gap. The notes in the two tone-scrambles were presented in different, random orders. In a "same"-stimulus, the two tone-scrambles were either both minor (target note = $B\flat_5$) or both major (target note = $B\natural_5$); in a "different"-stimulus, one tone-scramble was major and the other was minor. The participant indicated whether the two stimuli were the same type or different types by pressing a key or tapping an icon (see Sec. 2.4) and received feedback. The MMSD task comprised 16, randomly ordered trials, including four each of major-major, minor-minor, minor-major and major-minor tone-scramble pairs.

2.2.4 The wildcard categorization task

The subject was randomly assigned to one of 7 task conditions: the 1v2- 2v3-, 4v5-, 5v6-, 8v9-, 9v10- or the 10v11-task. Each of these tasks was analogous to the MMC task, except using different target notes. We code the notes of the (equal-tempered) chromatic scale between G_5 and G_6 as follows: $A\flat_5 \to 1$, $A_5 \to 2$, $B\flat_5 \to 3$, $B_5 \to 4$, $C_6 \to 5$, $C\sharp_6 \to 6$, $D_6 \to 7$, $E\flat_6 \to 8$, $E_6 \to 9$, $F_6 \to 10$, $F\sharp_6 \to 11$. The two target notes in a given wildcard task are the two notes whose codes appear in the name of the task. For example, in the 1v2-task, the target notes are 1 ($A\flat_5$) and 2 (A_5).

2.3 Self-report items

Before and after completing the three tasks, participants completed self-report items concerning demographics, musical training, and their current and past engagement with musical activities (see SI Text B for a full list of items).

2.4 User interface

The user-interface of the behavioral experiment appeared in the lower half of the screen while visual elements of a corresponding video game updated with the participant's progress in the upper half of the screen (Fig. S1, Supplementary materials). Subjects playing with a keyboard and/or mouse used the "F" and "J" keys to label each stimulus as Type-1 and Type-2, with clickable images shown on screen to match. Subjects with touch-screen devices instead used on-screen buttons labeled "1" and "2". The mapping of keys (or on-screen buttons) to the Type-1 and Type-2 stimuli was randomized for each subject on each task. During the MMSD task, images used to represent the keys/buttons were replaced by images bearing the symbols "=" and "≠", matching the randomized mapping. Feedback was provided as the message "CORRECT" or "INCORRECT" appearing immediately after each response in the lower half of the screen with corresponding video-game animations in the upper half. The feedback remained on-screen for 700 ms. This was followed by a 150 ms post-trial gap after which the stimulus of the next trial began to play automatically.

During the game, a bird avatar moved along an invisible grid, with the avatar's horizontal and vertical position corresponding, respectively, to the player's progress through the task and cumulative performance. The display provided subjects with their current streak (i.e., current number of consecutive correct responses, reset to 0 at the start of each task) and number of points, a value updated cumulatively over all three tasks based on correctness and response time. The points system was intended to motivate subjects to complete the game quickly but only insofar as their ability would allow. The reward in points for a correct response was equal to the punishment for an incorrect response of similar response time. See Supplementary materials Sec. S1.3 for details about point awards.

Upon completion, the participant was presented with a visualization of their score and told whether they had achieved "super-listener" status.

3. Results

3.1 Performance in the major-minor categorization task

The histogram of %-correct in the MMC is shown in Fig. 1. In line with previous findings, the histogram of %-correct in the MMC task is strongly bimodal: as Table 3 shows, the null hypothesis that %-correct is unimodally distributed in the MMC task is rejected with $p < 0.0001$ by Hartigan's dip test (Hartigan & Hartigan, 1985). Also shown in Table 3 (as well as in Tables 1, 2, and 4) are N, the number of subjects contributing to the histogram; *85%-or-more*, the proportion who responded correctly on at least 85% of the trials; and *dip*, the statistic computed in Hartigan's dip test (See Supplementary materials Sec. S2 for details). 85%-or-more provides a rough measure of the proportion of super-listeners contributing to the histogram, and Hartigan's dip statistic reflects the relative strength of the deviation from unimodality. Thus, for example, two histograms which both have dip-test-p < 0.0001 may differ in the strength of their deviation from unimodality; this difference will be reflected by the difference between their dip statistics.

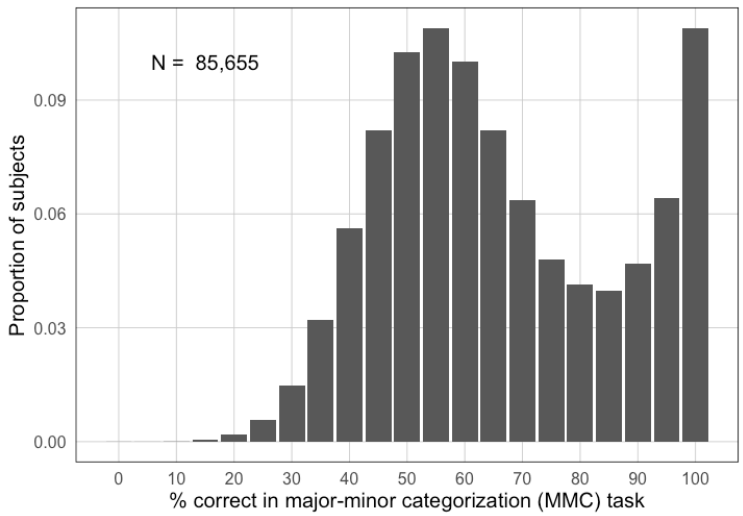

Figure 1: Histogram of percent correct in the major-minor categorization (MMC) task across all subjects.

If we identify a subject's *skill* in the MMC task as the expectation of their %-correct, then we face the following question: across our sample of subjects, what is the distribution of skills? A detailed answer to this question is beyond the scope of this paper; however, several observations may be useful. First, it is unlikely that the actual skill of any subject is below 50%; more plausibly, the many subjects who achieved scores below 50% were non-supers who happened by chance to score lower than their skills in the 20 trials of the MMC task. Of course, it is also possible for non-supers to get lucky and score higher than their skills. On the other hand, a super-listener with ability equal to 100% is guaranteed to score 100%. As this observation suggests, super-listeners are unlikely to contribute to the lower mode (at 55%) of the histogram in Fig. 1; however, it is likely that some non-supers contribute to the higher mode (at 100%). More generally, we can think of the histogram in Fig. 1 as a transformation that blurs the underlying distribution of skills strongly near 50% and preserves it more accurately for skills near 100%.

As these reflections suggest, the underlying distribution of skills probably has two cusps, a large one (due to non-supers) that decreases sharply from 50% and another smaller one (due to super-listeners) that increases sharply to 100%. Although our emphasis is on subjects whose skills are concentrated in these two cusps, there also probably exists a low concentration of subjects whose skills fall along the continuum between these two cusps.

Table 1: Histogram statistics for the MMC task across the 20 countries with the most subjects

Country	N	85%-or-more	dip	dip-test-p
United States	33,764	0.2746	0.0310	<0.0001
Canada	5,473	0.2498	0.0254	<0.0001
United Kingdom	4,616	0.2879	0.0321	<0.0001
Germany	3,200	0.2694	0.0295	<0.0001
Australia	2,617	0.3049	0.0375	<0.0001
Spain	2,242	0.2498	0.0229	<0.0001
Argentina	2,065	0.2344	0.0248	<0.0001
Mexico	1,777	0.2054	0.0149	0.0095
France	1,728	0.2604	0.0302	<0.0001
China	1,458	0.2634	0.0392	<0.0001
Russia	1,374	0.1870	0.0220	0.0001
Turkey	1,346	0.1062	0.0070	0.9739
Chile	1,100	0.3000	0.0333	<0.0001
Sweden	1,073	0.2665	0.0344	<0.0001
Italy	1,029	0.2634	0.0371	<0.0001
The Netherlands	999	0.2392	0.0266	<0.0001
India	983	0.1129	0.0126	0.3984
Brazil	866	0.1778	0.0110	0.7533
Poland	817	0.2375	0.0235	0.0037
Colombia	765	0.1987	0.0209	0.0223

3.2 The bimodality persists across participants of different backgrounds

Table 1 shows the statistics for histograms of %-correct in the MMC task from the 20 most common countries in our data set. For more than half of these conuntries dip-test-$p < 0.0001$, and in all but 3 cases, the dip-test-p is less than 0.02. Thus, performance in the MMC task tends to be bimodally distributed across most countries. The histograms for Brazil, India and Turkey all yielded dip-test-$p > 0.5$ indicating that they were closer to unimodal than to bimodal. Not surprisingly, these are also the three countries with the lowest proportions of subjects who scored 85%-or-more, suggesting that the samples from these countries comprised unusually low proportions of super-listeners.

Table 2 shows analogous results for the 20 most common native languages in our data set. In 13 cases, $p < 0.0001$. For all but 3 native languanges (Turkish, Hindi and Tagalog), the histogram was closer to bimodal than unimodal. Typically, high p-values in Hartigan's dip test result because the high-performance mode is too small. The reverse is true for the sample of native Korean speakers: the proportion of native Korean speakers who scored 85% or more correct was unusually high: 0.4631. By contrast, for example, the proportion of native Hindi speakers who scored 85% or more correct was 0.0808. It would be a mistake, however, to ascribe much significance to these variations. The word-of-mouth process that generates the sample of native speakers of a given language may well spread through a subgroup all of whom tend to have the same level of musical competence; by chance, this may be high for some languages and low for others.

Fig. 2 shows the histograms of %-correct in the MMC task for native speakers of tonal, pitch-accented and other languages. There are slight differences between the histograms for these different classes. In particular, as shown in the figure, proportion of subjects with 85%-or-more correct is 0.2950 for native speakers of tonal languages, 0.2813 for native speakers of pitch-accented languages and 0.2580 for native speakers of languages that fall in neither of these categories. It is possible that these differences reflect systematic differences in sensitivity to tone-scrambles conferred by early linguistic experience. However, it also possible that these differences are due to the vagaries of our sampling methods. Importantly, each of these three classes of language comprises a large subgroup of non-supers, implying that early language exposure exerts little, if any, influence on super-listenerhood.

Table 2: Histogram statistics for the MMC task across the 20 native languages with the most subjsects

Native language	N	85%-or-more	dip	dip-test-p
English	46,623	0.2782	0.0308	<0.0001
Spanish	9,407	0.2264	0.0209	<0.0001
German	3,455	0.2715	0.0311	<0.0001
French	2,371	0.2522	0.0262	<0.0001
Russian	1,908	0.1950	0.0225	<0.0001
Chinese/Mandarin	1,832	0.3308	0.0564	<0.0001
Turkish	1,441	0.1110	0.0070	0.9595
Portuguese	1,389	0.2153	0.0175	0.0051
Italian	1,077	0.2516	0.0320	<0.0001
Dutch	1,075	0.2372	0.0209	0.0027
Swedish	1,022	0.2769	0.0396	<0.0001
Polish	909	0.2299	0.0230	0.0017
Chinese/Cantonese/Yue	872	0.3727	0.0428	<0.0001
Finnish	580	0.3138	0.0414	<0.0001
Norwegian	509	0.3242	0.0429	<0.0001
Indonesian	477	0.2893	0.0477	<0.0001
Greek	429	0.2587	0.0374	0.0004
Hindi	421	0.0808	0.0185	0.4400
Tagalog	417	0.1127	0.0138	0.9037
Korean	406	0.4631	0.0258	0.0551

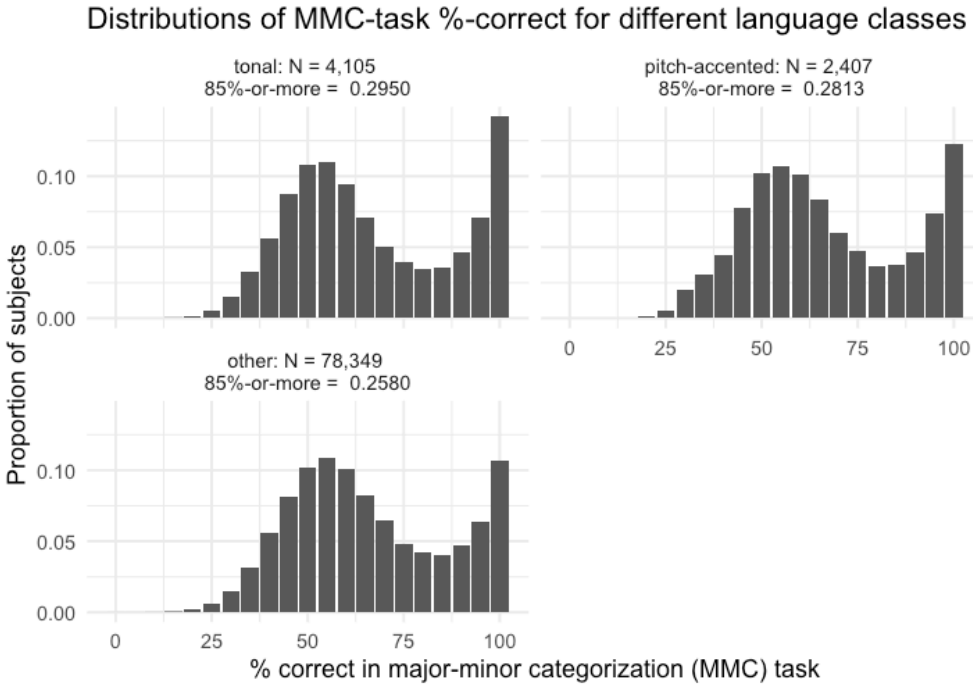

Figure 2: Histograms of %-correct in the MMC task for subjects with different classes of native languages.

Table 3: Histogram statistics for all tasks

Task	N	85%-or-more	dip	dip-test-p
1v2	11,181	0.2007	0.0056	0.0210
2v3	11,355	0.1939	0.0050	0.0580
MMC (3v4)	85,655	0.2599	0.0284	<0.0001
4v5	11,318	0.2100	0.0067	0.0015
5v6	11,482	0.1758	0.0034	0.5798
8v9	10,912	0.2470	0.0104	<0.0001
9v10	11,737	0.1390	0.0024	0.9766
10v11	11,639	0.1309	0.0019	0.9999
MMSD	85,655	0.1616	0.0007	0.9991

Table 4: Histogram statistics for the MMC task across groups with different amounts of music lessons

Years lessons	N	85%-or-more	dip	dip-test-p
Never had lessons	27,761	0.1042	0.0051	<0.0001
< 1 year	8,634	0.1893	0.0094	0.0001
1 to 2 years	9,959	0.2469	0.0176	<0.0001
2 to 4 years	9,818	0.3218	0.0322	<0.0001
4 to 6 years	11,259	0.4425	0.0153	<0.0001
6 to 10 years	6,188	0.1682	0.0106	<0.0001
> 10 years	9,600	0.5803	0.0063	0.0127

3.3 The association between musical training and MMC-task %-correct

As observed in previous lab experiments, the current results show that years-of-music-lessons is positively correlated with %-correct in the MMC task across the 55,458 subjects who provided this information (Fig. 3). It is very clear, however, that this correlation is due to two separate clusters of subjects: a cluster of non-supers whose mode is at 1-2 years-of-music-lessons and 55% correct, and a separate cluster of super-listeners whose mode is at > 10 years-of-music-lessons and 100% correct. A linear correlation coefficient is therefore not an appropriate measure of the strength of this association. Instead, we use the Spearman rank correlation ρ, which measures how well MMC task-score can be approximated by an arbitrary monotonic function of years-of-music-lessons. We find $\rho = 0.3037$ (p infinitesimal). We also note that (1) each of the two clusters of subjects is broadly elongated across years-of-music-lessons, and (2) a bimodal distribution in %-correct persists across all levels of musical training (see Table 4).

As shown in Fig. 4, the distribution of %-correct in the MMC task is also bimodally distributed across the subset of subjects who reported never having taken music lessons. Although this distribution has a large mass of subjects who perform near chance, roughly 6% of these subjects achieve 90% or more. That this distribution is not unimodal is confirmed by Hartigan's dip test ($p = <0.0001$).

3.4 Performance in the wildcard tasks

Figure 5 shows the histograms of %-correct in the wildcard tasks. Performance tends to be worse in these tasks than in the MMC task. Also, the histograms in Fig. 5 seem to be less strongly bimodal than that of the MMC task. This impression is confirmed by Hartigan's dip test (Table 3). The only wildcard task to yield a strongly bimodal distribution of %-correct was the 8v9-task. This is striking because the target notes in the 8v9-task are $E\flat_5$ (degree 6 of the G natural and harmonic minor scales, as well as of the descending G melodic minor scale) and $E\natural_5$ (degree 6 of the G-major scale). After degree 3, degree 6 is generally held to be the next most important in determining the majorness vs minorness of the scale.

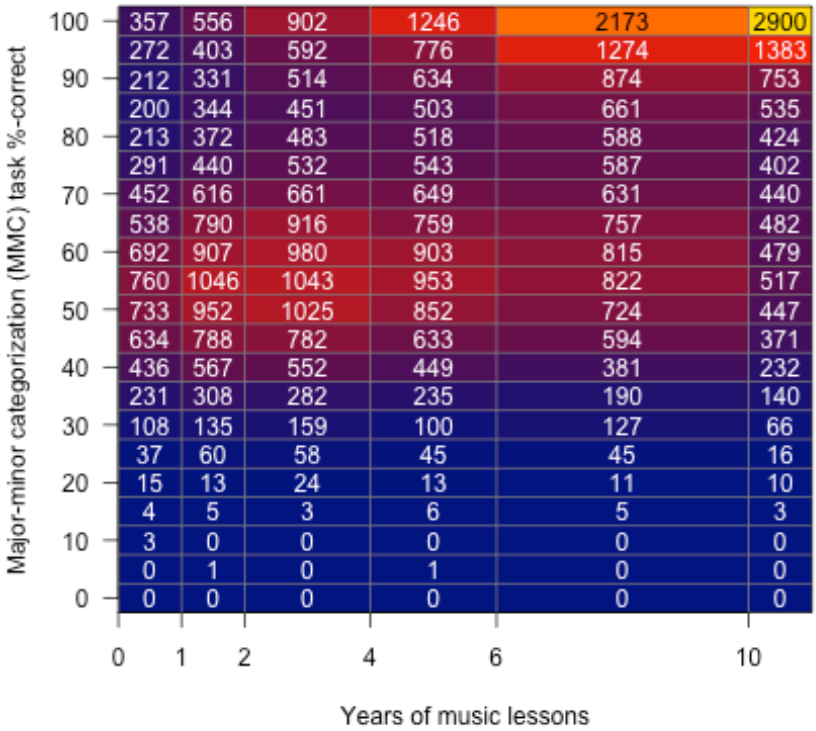

Figure 3: A heat map representing the joint distribution of subjects' scores on the MMC task and their years of musical training. The numbers (and color) within each cell represent the number of subjects who achieved the corresponding score and reported the corresponding years of training. Note that the bins along the x-axis correspond to the fixed responses available to participants: "Less than a year", "1-2 years", "2-4 years", "4-6 years", "6-10 years", and "More than 10 years".

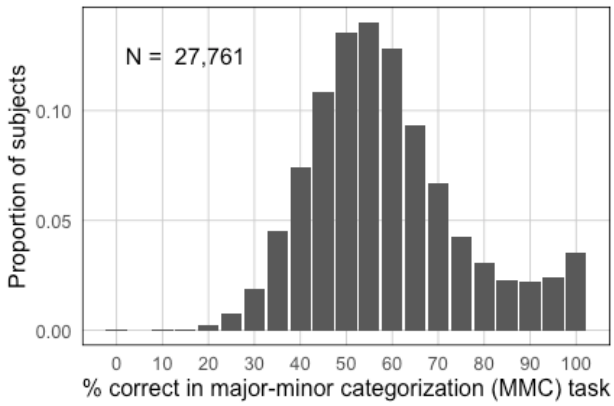

Figure 4: Histogram of percent correct in the MMC task achieved by subjects who reported never having taken music lessons.

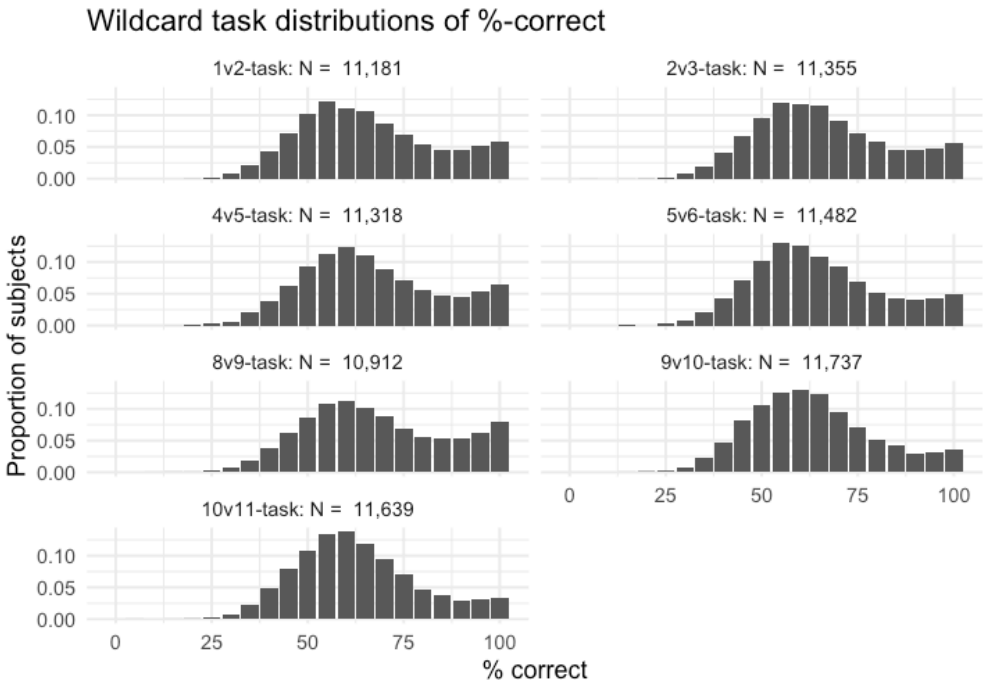

Figure 5: Histograms of %-correct in the wildcard tone-scramble tasks.

3.5 Performance in the major-minor same-different (MMSD) task

We were initially surprised by the results from the MMSD task. In striking contrast to the MMC task, the MMSD task (Fig. 6, Table 3) yields strongly unimodal performance. One reason for this is that many subjects who performed perfectly in the MMC task failed to do so in the MMSD task: around 11% of subjects scored 100% in the MMC task compared to only 4% in the MMSD task. The Supplementary material discusses possible reasons why the MMSD task is more difficult for super-listeners than the MMC task. The Supplementary materials also include an analysis showing that the MMSD task is very slightly easier than the MMC task for non-supers; however, the size of this effect is negligible, suggesting that whatever labeling problem non-supers may experience is not an important source of their difficulty in performing the MMC task. We conclude that non-supers discern little, if any, difference between major vs minor tone-scrambles.

4. Discussion

Super-listeners evidently possess some type of auditory sensitivity, which we call "MM-sensitivity", that non-supers lack. What is the nature of MM-sensitivity?

To address this question, we focus on Fig. 3, which makes several things clear. First, for many non-supers, music lessons do not increase MM-sensitivity. This is evident when we consider the horizontally elongated ridge of subjects whose mode is at years-of-lessons = 2 and %-correct = 55. The subjects in this ridge must have had low MM-sensitivity when they started taking lessons, and whatever lessons they took failed to boost their MM-sensitivity. Second, many other subjects who took few (or no) years of lessons possess high MM-sensitivity. Thus, music lessons are neither necessary nor sufficient for high MM-sensitivity.

Consider the large group of super-listeners who took many years of lessons. How did these "highly-trained super-listeners" attain their high MM-sensitivity? Many probably had high MM-sensitivity before starting lessons. Indeed, their high MM-sensitivity may have been instrumental in prompting these subjects to seek out lessons and to persist in taking them once they had started. On the other hand, the fact that music

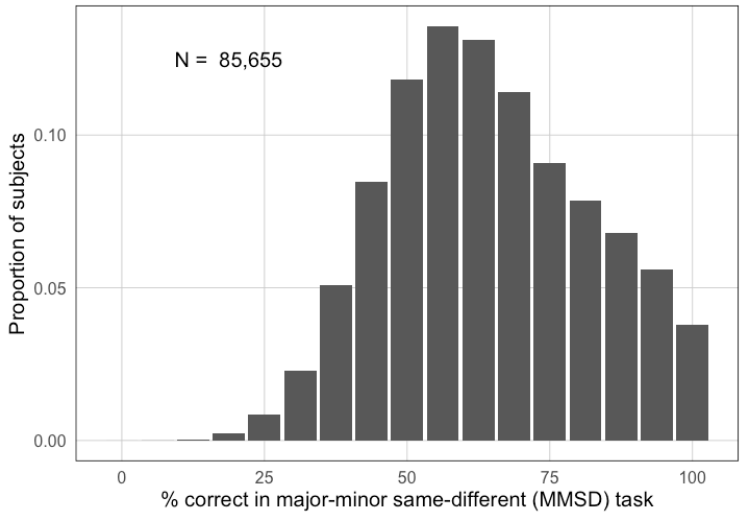

Figure 6: Histogram of percent correct in the MMSD task across all subjects.

lessons fail to boost MM-sensitivity for many non-supers raises the possibility that this may be true for all non-supers. If so, then all highly-trained super-listeners would have had high MM-sensitivity before starting lessons. Our data are consistent with this possibility.

We cannot rule out the possibility that some highly-trained super-listeners may have started out as non-supers who were transformed into super-listeners by the lessons they took; however, our data suggest that such subjects are rare. It is easy to imagine that the extended exposure to Western music that a non-super receives in taking many years of music lessons might gradually increase MM-sensitivity to super-listener-level. The following scenario seems less plausible: after taking music lessons for some number of years, a non-super is abruptly transformed into a super-listener by a sudden jump in MM-sensitivity. We assume that the second scenario occurs rarely, if at all. However, if it were common for subjects to start out as non-supers and transform gradually into super-listeners by taking music lessons, then we should see elevated concentrations of subjects in one or more of the rectangles in Fig. 3 corresponding to years-of-music-lessons intermediate between 0 and > 10 and %-correct intermediate between 50 and 100. We see no sign of such an effect. This suggests that a large majority of the highly-trained super-listeners in our sample probably had high MM-sensitivity before they began music lessons. We note, however, that this inference is speculative; a longitudinal study in which children are tested in the MMC task before starting music lessons and at intervals thereafter is required to test it.

Super-listeners tend to take more years of lessons than non-supers. Across the subjects who reported having taken music lessons, if we assume that those who responded "Less than a year", "1 to 2 years", "2 to 4 years", "4 to 6 years", "6 to 10 years" and "More than 10 years" took (respectively, on average) 0.5, 1.5, 3, 5, 8 and 11 years of lessons and take all subjects who scored 95% or 100% as super-listeners and all who score below 70% as non-supers, we find that the mean number of years of lessons is 7.09 for super-listeners but only 4.28 for non-supers.

Unquestionably, however, musical proficiency tends to increase with years of music lessons; thus, if most of the highly-trained super-listeners in our sample had high MM-sensitivity before they started lessons, then another implication of Fig. 3 is that super-listeners tend to attain greater musical proficiency than non-supers.

Do super-listeners experience music differently from non-supers? Super-listeners report that major and minor tone-scrambles possess qualities that mimic the "happiness" and "sadness" of music in the major and minor scales. This might tempt us to conclude super-listeners can sense scale-derived qualities in music, whereas non-supers cannot. However, this conclusion is clearly too strong: several studies confirm that nearly all subjects are sensitive to scale variations in carefully structured examples of actual music (Temperley & Tan,

2013; Bonetti & Costa, 2019). The possibility remains, however, that super-listeners are more sensitive to scale-derived qualities than non-supers.

Is it possible that non-supers are just as sensitive as super-listeners to scale-derived qualities in real music but cannot extract them from tone-scrambles? There are several reasons to doubt this. First, the finding that super-listeners tend to attain higher levels of musical proficiency suggests that super-listeners are more sensitive than non-supers to *some* qualities in real music, and scale-derived qualities are the most likely candidates. In addition, several studies have shown that modifying tone-scrambles to make them more like real music does not help non-supers: non-supers show no improvement either when tone-scrambles are presented at rates typical of actual music (Mednicoff et al., 2018), or when various sorts of sequential structure are added to them (Ho & Chubb, 2020).

One might object that many studies have shown that subjects do, in fact, tend to hear major music as "happy" and minor music as "sad." Although this is true, nearly all of these studies (Temperley & Tan (2013) and Bonetti & Costa (2019) are exceptions) have reported modest mean effects consistent with the bimodal distribution reported here. Other studies have found that subjects are surprisingly poor at discriminating between major vs minor music (Halpern, 1984; Halpern et al., 1998). Finally, experiments requiring subjects to classify triadic chords as major or minor have observed bimodal performance similar to the current results (Blechner, 1977; Crowder, 1985).

In summary, the current results lead us to the working hypothesis that super-listeners are more sensitive than non-supers to scale-defined qualities in actual music.

5. Contributions

S. Waz: Conceptualization, Data curation, Investigation, Methodology, Software, Validation, Writing–review & editing. S. Mednicof: Conceptualization, Investigation, Methodology, Writing–review & editing. J. Ho: Conceptualization, Investigation, Methodology, Writing–review & editing. C. Hilton: Data curation, Software, Writing–review & editing. G. Hickok: Conceptualization, Funding acquisition, Methodology, Writing–review & editing. S. Mehr: Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Writing–review & editing. C. Chubb: Conceptualization, Formal analysis, Methodology, Software, Writing–original draft.

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